Revolutionizing 5G/6G Patch Antenna Design with Machine Learning

PUTSHU LUNGHE Samuel, *samuellunghe@gmail.com ISBAT University Kampala*

1. ABSTRACT

Traditional antenna design is slow and labor-intensive, often taking weeks of manual tuning with suboptimal results due to complex parameter interactions. Our machine learning-powered solution delivers optimized microstrip patch designs in under 60 seconds a big speedup while improving frequency accuracy, bandwidth, and return loss by 33% on average, validated against 1,648 professional simulations. This breakthrough democratizes advanced 5G/6G patch antenna design, providing engineers and researchers with an intuitive web interface that redefines how antennas are developed and optimized.

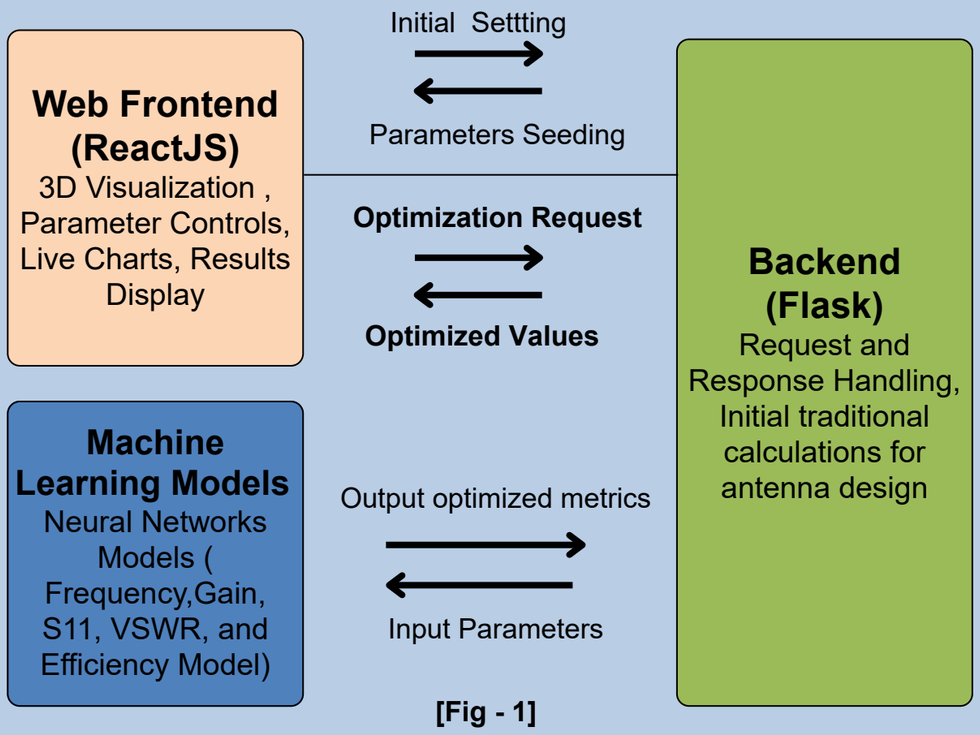
2. INTRODUCTION

The rapid rise of 5G and the coming of 6G demand advanced antennas with strict performance requirements, yet traditional microstrip patch design remains slow and limited. Conventional methods take 3-4 weeks of trial-and-error simulations, each lasting up to two hours, and require specialized RF expertise. Even after extensive effort, engineers often settle for suboptimal trade-offs between frequency accuracy, bandwidth, gain, and efficiency, delaying product development and increasing costs in competitive markets.

Our **machine learning-driven system** transforms this process by generating optimized designs in seconds rather than weeks. Using neural networks trained on **1,648 professional simulations**, it captures complex electromagnetic behavior and delivers antennas with consistently superior performance. The platform balances multiple objectives simultaneously, improving frequency accuracy, bandwidth, and return loss by over 30%. Through an intuitive web-based 3D interface, this breakthrough makes high-performance 5G/6G antenna design faster, smarter, and accessible to engineers and researchers at all levels.

3 . METHODOLOGY

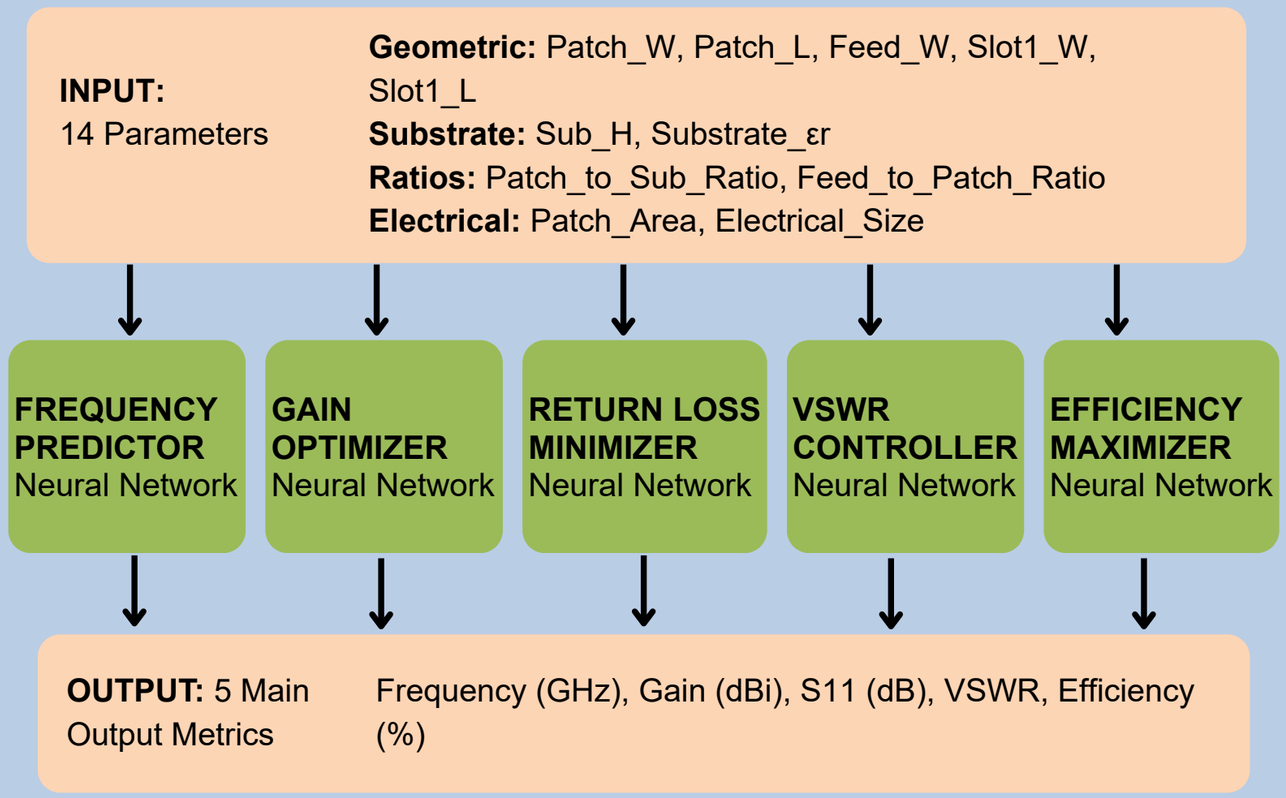
a. System Architecture Overview



[Fig - 1]

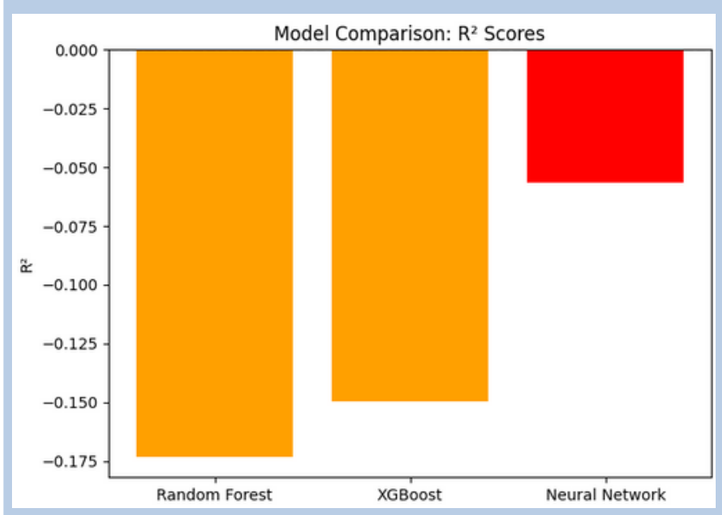
Our approach employs an ensemble of neural networks trained on **1,648 professional antenna simulations,** using **14 key geometric,** substrate, slot, and electrical parameters to predict frequency, gain, return loss, VSWR, and efficiency with high accuracy. A genetic algorithm integrated with the AI models enables rapid optimization, exploring thousands of design variations in real time and delivering predictions in under 100 ms compared to the 30+ minutes required by conventional simulations. The system is deployed as a web-based platform, combining a React frontend with interactive 3D visualization and a Flask backend for model serving and optimization, creating an accessible, fast, and data-driven antenna design workflow.

b. AI Model Pipeline



[Fig - 2]

4. RESULTS

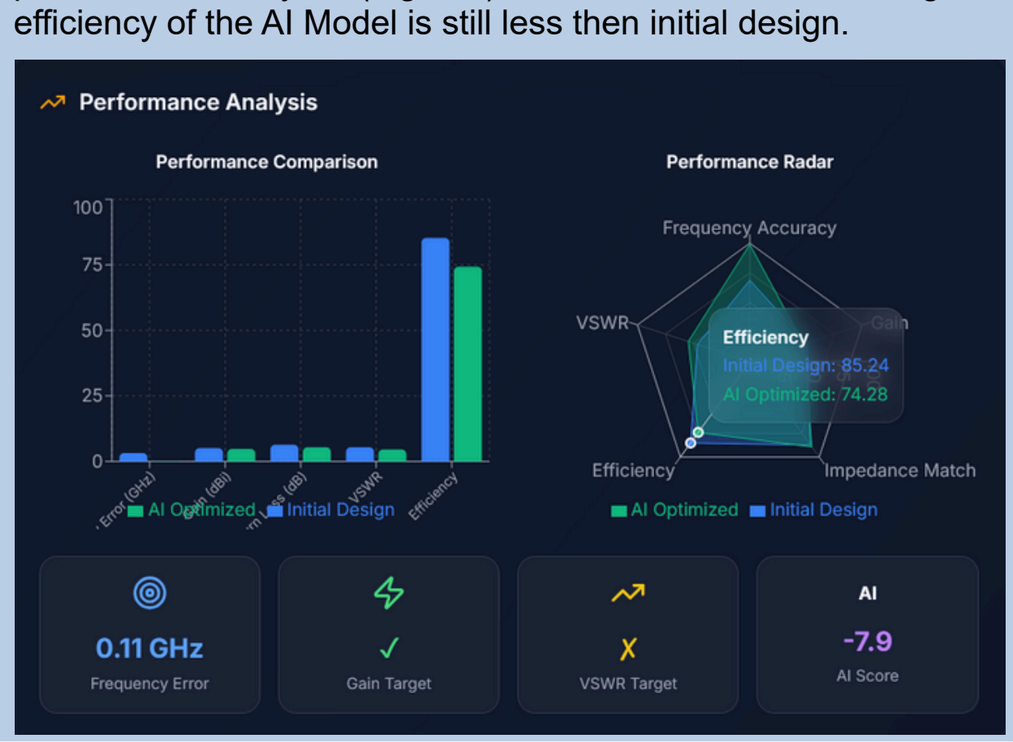


Machine Learning Models Evaluation

Model	MAE	R2
Random Forest	16.099736	-0.173282
XGBoost	15.949372	-0.149571
Neural Network	15.137690	-0.039284

[Fig - 3]

Neural Network achieved the lowest MAE (~15.47), but all models have negative R^2, showing **underfitting**. This suggests that dataset complexity and feature engineering need improvement for better predictive performance. As we can see on the perfomance analysis (Fig - 4) on the frontend for a target of 28GHz design, the efficiency of the AI Model is still less then initial design.



[Fig - 4]

5. CONCLUSION

Although the neural network achieved the lowest MAE (~15.47), the negative R^2 values across all models indicate underfitting. This highlights the need for richer datasets and improved feature engineering to enhance predictive performance.

6. REFERENCES

J. Colaco and R. Lohani, "Design and Implementation of Microstrip Patch Antenna for 5G applications," 2020 5th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2020, pp. 682-685, doi: 10.1109/ICCES48766.2020.9137921

S. K. Goudos, P. D. Diamantoulakis, M. A. Matin, P. Sarigiannidis, S. Wan and G. K. Karagiannidis, "Design of Antennas through Artificial Intelligence: State of the Art and Challenges," in IEEE Communications Magazine, vol. 60, no. 12, pp. 96-102, December 2022, doi: 10.1109/MCOM.006.2200124.